Artificial Intelligence in Veterinary Medicine: From Laboratory Accuracy to Clinical Readiness — Secondary Analysis of 39 Deep-Learning Studies, a Real-World Benefit–Harm Appraisal, and the Veterinary Translational Readiness Index (VTRI)

Abdenasser Mahdjoubi, DVM*

Faculty of Veterinary Medicine, Ibn Khaldoun University, Algeria (graduate)

*Corresponding author: dr.mahdjoubi@gmail.com | ORCID: 0009-0007-9528-7049

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Abstract

Background. Artificial intelligence (AI) is moving rapidly from proof-of-concept to routine veterinary use, yet published accuracy figures come mostly from controlled studies, while independent evaluations of commercial tools in general practice are few. No analysis has set laboratory performance, real-world performance and the maturity of the evidence side by side.

Methods. We anchored the review on a recent PRISMA-based systematic review of deep learning in veterinary diagnostics (422 PubMed records screened, 39 primary studies, 2013-2024) and performed a secondary quantitative analysis of those 39 studies (modality, species, sample size). We added targeted 2019-2026 evidence on commercial radiology AI, AI devices, large language models (LLMs), clinical documentation tools, adoption surveys, market reports and regulatory statements, and classified each figure by evidence type (peer-reviewed, independent, vendor-reported, projection). We developed the Veterinary Translational Readiness Index (VTRI), a five-domain, 15-point rubric of evidence maturity, paired with a three-level Real-World Performance (RWP) flag.

Results. Radiography and cytology each account for 33% of studies; 82.1% addressed dogs or cats and 10.3% livestock or horses. Median sample size was 837.5 (IQR 275-2,355); 52.6% of studies used fewer than 1,000 cases, and datasets were not larger after 2021. None of the 39 studies addressed interpretability technically. Laboratory accuracy of 82.7-99% contrasted with independent evaluations of six commercial platforms on canine abdominal radiographs (sensitivity 28-78%; small-intestinal obstruction 23-69%), a commercial pulmonary-nodule tool (sensitivity 55.4%), an AI stethoscope (atrial-fibrillation PPV 21.4%) and AI triage (about 60% of non-urgent cases over-triaged). Reported adoption rose from 39.2% (2024) to 83.7% (2026) in North American surveys, while specialty colleges stated that no commercial imaging product met transparency, validation and safety standards. VTRI scores ranged from 2 to 7 of 15; interpretability scored 0 in every class.

Conclusions. AI is beneficial in narrow, well-labelled tasks, but adoption has outpaced independent validation, interpretability and regulation. Deployment should be clinician-supervised decision support with external validation, prevalence-aware reporting and local validation before importation, particularly in low-resource regions.

Keywords: artificial intelligence; deep learning; veterinary medicine; diagnostic imaging; external validation; large language models; regulation; translational readiness

1. Introduction

In human medicine, deep-learning systems perform at or near clinician level on several image-classification tasks [12,13]. Veterinary medicine is following, later and from a weaker data position. A 2025 systematic review concluded that deep learning in veterinary diagnostics remains at an early stage, limited by small and heterogeneous datasets, narrow species coverage and the absence of regulatory validation pathways [1]. Overviews and narrative reviews reach similar conclusions [14,16,18-20,22,40].

Two literatures have developed side by side. The first is a research literature of algorithm studies reporting high accuracy, largely on curated, single-centre data [1]. The second is a practice-facing literature of commercial products, vendor-reported efficiency gains, adoption surveys, market reports and position statements from professional bodies [42-44,48-61]. Between them sits an evidence gap: when commercial tools are tested independently on general-practice data, results are often markedly worse than the headline figures [23,24,27]. At the same time, reported use of AI by veterinarians has more than doubled in two years [48,49], and an audit of 71 commercial veterinary AI products found that nearly two-thirds of vendors disclosed no validation metric [21].

Existing work addresses parts of this picture: catalogues of applications [18-20,40], transparency of commercial products [21], ethics [37] and regulation [42-44]. To our knowledge, and within the limits of a targeted (non-systematic) search conducted in October 2026, no published work (i) quantifies the scale and species coverage of the 39 deep-learning diagnostic studies in the most recent PRISMA-based review, (ii) sets laboratory performance against independent real-world evaluations in one comparison, and (iii) scores the maturity of the evidence by application class with an explicit, contestable rubric.

This paper therefore makes four contributions: (1) a secondary quantitative analysis of the 39 studies in the review of Xiao et al. [1]; (2) a structured lab-to-clinic comparison of laboratory and independent real-world performance, with every figure labelled by evidence type; (3) an integrated benefit-harm appraisal covering commercial radiology AI, devices, LLMs, documentation tools, governance and cost; and (4) the Veterinary Translational Readiness Index (VTRI) with a Real-World Performance flag, proposed as a falsifiable rubric complementing transparency tools such as VATI [21], together with a perspective on validation in low-resource regions, including Algeria.

2. Methods

2.1 Design

This is a review with secondary quantitative analysis. It is not a new PRISMA systematic review and makes no claim of exhaustive retrieval. It reports no new animal or human data.

2.2 Evidence base

The backbone is Xiao et al. [1], which searched PubMed for "deep learning" and "veterinary", identified 422 records, excluded 66 non-primary articles, 142 articles that mislabelled simple networks or used deep learning only for pre-processing, segmentation or non-diagnostic purposes, and 174 human-medicine articles plus one duplicate, and retained 39 primary studies (2013-2024) [1,15]. Targeted searches (October 2026) added evidence on commercial radiology AI [2,3,23,27], devices and triage [24,25], oncology tools [35,36], LLMs [8-11,63], documentation tools [57,58], adoption surveys [48-50], market reports [51-56], pricing [59,61] and regulatory or professional statements [42-47].

2.3 Secondary extraction and analysis

From Table 1 of Xiao et al. [1] we extracted year, modality, species and reported sample size for each of the 39 studies (Appendix A). Sample sizes are reported in different units (images, animals, slides, records) and are treated as descriptive indicators of dataset scale, not comparable statistical power. Medians, interquartile ranges and Mann-Whitney U tests (two-sided, alpha = 0.05) were computed in Python (SciPy). Performance metrics were not pooled because tasks, metrics and reference standards are not commensurable; they are tabulated as reported.

2.4 Classification of evidence type

Every performance, adoption, efficiency or market figure was assigned one of four labels: [P] peer-reviewed measured result; [I] independent external evaluation of a commercial product; [V] vendor- or company-reported, not independently verified; [F] forecast or projection. Labels appear in the tables and are used when comparing laboratory and real-world results.

2.5 VTRI development

Five domains were derived from limitations repeatedly identified in the source literature [1,3,9,23,42]: D1 data adequacy; D2 validation level; D3 interpretability; D4 species and population coverage; D5 clinical and regulatory integration. Each is scored 0-3 (Table 5). Because a maturity score can rate a tool well even when its measured performance is poor, each class also carries a Real-World Performance (RWP) flag: Green = independent real-world data concordant with laboratory results; Amber = no or limited independent real-world data; Red = independent data show clinically important deficits. Scores and flags were assigned by the author from the evidence cited here. They are preliminary expert judgements, not measurements, intended to be contested and refined through a modified Delphi process (Section 9).

3. Results

3.1 Landscape of the evidence

Between 2013 and 2024 the 39 included studies were distributed as 11 (2013-2019), 13 (2020-2021) and 15 (2022-2024), indicating steady growth rather than an inflection. Radiography and cytology each represented 33% of studies, followed by health-record analysis (8%), MRI (8%), environmental data (5%), photo/video (5%) and ultrasound (5%) [1]. Convolutional neural networks remained the dominant architecture, and the review observed no

clear upward trend in accuracy over time; performance depended on task, species, dataset size and model complexity [1].

3.2 Species and dataset scale (secondary analysis)

In our extraction, 32 of 39 studies (82.1%) concerned dogs, cats or both, 4 (10.3%) concerned cattle, pigs or horses, 2 used rodents and 1 did not specify species. (The source review reports 84% for dogs and cats under its own classification [1]; the small difference reflects how mixed-species studies were assigned.) Sample size was reported for 38 studies: median 837.5 (IQR 275-2,355), minimum 32, maximum 364,249. Twenty studies (52.6%) used fewer than 1,000 cases, 17 (44.7%) used 500 or fewer, and only 4 used 10,000 or more.

Dataset scale differed by modality. Radiograph studies (n = 13) had a median of 1,000 (range 60-22,120), cytology/histology studies (n = 13) a median of 352 (32-3,845), and sensor/EHR/text studies (n = 4) a median of 63,950 (1,000-364,249). The radiograph-versus-cytology/histology difference was not statistically significant (Mann-Whitney U = 115, p = 0.124), and studies from 2021 onward (n = 22, median 550) were not larger than those up to 2020 (n = 16, median 1,023; U = 199, p = 0.51). The field has therefore not moved toward larger datasets even as architectures matured, consistent with the review's conclusion that dataset creation is the principal bottleneck [1]. These tests are exploratory, use small groups and heterogeneous units, and should be read as descriptive.

3.3 Diagnostic imaging

Thoracic radiography. The largest study trained a network on 22,120 canine and feline radiographs to identify 15 primary thoracic lesion types and reported an overall AI error rate of 10.7%, significantly lower than that of veterinarians working alone or with AI output; veterinarians shown the AI output did not improve significantly [1,4]. For cardiac size, a network agreed with specialists on vertebral heart scale with an intraclass correlation of 0.998 (n = 60) [5], and a separate vertebral-heart-score algorithm has been diagnostically validated [62]; canine cardiomegaly detection reached AUROC above 0.9 [6]. A convolutional network detecting left atrial enlargement in 792 lateral radiographs reached 82.71% accuracy, 68.42% sensitivity and 87.09% specificity, comparable to board-certified radiologists in the same study [7]. Lung-lesion detection reached only 72.3-79.6% accuracy and tracheal-collapse grading 98.9% [1]. In feline radiographs, ResNet-50 and Inception V3 models reached AUC above 0.9 for alveolar, bronchial and pleural-effusion patterns [30]. Further classifiers for canine and feline thoracic and cardiac disease have been reported, including automatic classification of canine thoracic radiographs [29], comparison with radiologists for cardiogenic pulmonary oedema [31] and deep-learning diagnosis of feline hypertrophic cardiomyopathy [17]. The picture is not one-sided: four DenseNet models were compared with 13 board-certified radiologists across 15 canine thoracic labels, and the specialists had a statistically significantly lower error rate than all four models [28].

MRI, ultrasound and orthopaedics. A deep-learning model detected intervertebral disc disease on canine thoracolumbar MRI with sensitivity 0.908-1.0, but sensitivity for neoplasms ranged from 0 to 37.5% and for syringomyelia from 0 to 10% because training cases were scarce [1]. A hip-dysplasia model achieved specificity 0.92 but sensitivity only 0.53 on an unbalanced dataset [1]. Five-stage classification of canine chronic kidney disease from ultrasound reached average accuracy 0.46, attributed partly to ultrasound being an unreliable ground truth for staging [1]. These failures share a cause: rare classes and unreliable labels.

Commercial radiology AI (head-to-head). In a comparison of commercial software (SignalRAY/SignalPET) with 11 board-certified radiologists (ECVDI or ACVR) on 50 anonymised canine and feline studies, the software matched the best radiologist in accuracy, was more specific and was less sensitive than the radiologists [2]. A published commentary criticised the design: a small single-institution sample without a power calculation, class imbalance that rewards favouring the common normal output, and the lack of a true gold standard [3]. Independent evaluations in general practice are summarised in Section 4.

Table 1. Reported performance in veterinary diagnostic imaging (as reported; not pooled).

Task / species	Data	Reported result	Type	Ref.
Primary thoracic lesions (15 types), dogs and cats	22,120 radiographs	AI error 10.7%; lower than veterinarians; vets with AI output not significantly better	P	[1,4]
Vertebral heart scale, dogs and cats	60 radiographs	ICC 0.998 vs specialists	P	[1,5]
Canine cardiomegaly	1,465 radiographs	AUROC above 0.9	P	[1,6]
Canine left atrial enlargement	792 radiographs	Acc 82.71%; sens 68.42%; spec 87.09%	P	[7]
Lung lesions, dogs and cats	2,862 radiographs	Acc 72.3% (abnormal lung) / 79.6% (lesion detection)	P	[1]
Canine tracheal collapse grading	600 radiographs	Acc 98.9%; sens 98.3%; spec 99.2%	P	[1]
Four DenseNet models vs 13 radiologists, 15 labels	Canine thoracic radiographs	Specialists had significantly lower error rate	P	[28]

Task / species	Data	Reported result	Type	Ref.
Canine hip dysplasia (two-stage)	Unbalanced set	Spec 0.92; sens 0.53	P	[1]
Canine spinal MRI: IVDP / neoplasm / syringomyelia	2,693 MRI	IVDP sens 1.0; neoplasm sens 0-37.5%; syringomyelia 0-10%	P	[1]
Canine CKD, 5-stage ultrasound	883 ultrasound	Accuracy 0.46	P	[1]
Commercial AI vs 11 radiologists	50 studies	Matched best radiologist in accuracy; higher specificity; lower sensitivity	P	[2,3]

3.4 Cytology, histopathology and laboratory data

Digital pathology showed the strongest laboratory results. Reticulocyte classification in feline blood smears reached 98.7% accuracy (n = 1,046); a model scored equine exercise-induced pulmonary haemorrhage cytology at 92.3% accuracy against 76% for human experts (n = 52); seven canine skin-tumour classes were classified at about 95%, still below human expert level; and three canine lymphoma classes were separated with 99% accuracy in a preliminary test set (n = 1,530) [1]. A large-laboratory mitotic-figure system improved on human baselines, and the review noted that pathologists vary in the slide region they select, producing inconsistent mitotic counts [1]. More recent studies add nuance. A 2026 proof-of-concept network (Inception-v3) classified seven common canine skin tumours on whole-slide images with only 75.6-80.9% accuracy, and the authors cautioned that this likely overestimates routine performance [26]. A ResNet50 model separated canine lymphoma from reactive lymph-node hyperplasia with 95.4% accuracy and AUC 0.988 [33], and EfficientNet models exceeded 90% accuracy on clinical-image dermatology lesions (for example alopecia 98.12%) [34]. The SediVue Dx urine analyser has a published validation against manual microscopy (530 samples; sensitivity 85-90% and specificity 87-90% for red cells, white cells and struvite) [32]. The two skin-tumour figures (about 95% [1] versus 75.6-80.9% [26]) come from different studies with different datasets and should not be read as one model degrading; they show how wide the spread of reported performance is within a single task class. Cytology/histology studies had the smallest median dataset among the large modality groups (352), and several results are preliminary or single-centre [1].

3.5 Health records, language models and clinical documentation

EHR and text. A recurrent network applied to 106,251 feline records predicted early chronic kidney disease with sensitivity 0.907 and specificity 0.989, and an LSTM classified necropsy reports with F1 0.932 (gastrointestinal), 0.947 (respiratory) and 0.752 (urinary) [1]. GPT-4o extracted six clinical signs of feline chronic enteropathy from 250 EHRs with sensitivity 96.9%, specificity 97.6%, positive predictive value (PPV) 80.7% and balanced accuracy 96.3%; the authors considered it a viable alternative to manual extraction requiring setting-specific validation [8]. The gap between 97.6% specificity and 80.7% PPV shows that for uncommon findings roughly one in five positive extractions would still be wrong, a pattern to expect whenever prevalence is low.

LLMs in knowledge tasks. A University of Georgia study of 495 faculty-written questions reported that GPT-3.5 scored 55% and GPT-4 77%, both significantly below veterinary students (86%) [63] (as also summarised in a review [9]). A Hong Kong study found that performance declined with question difficulty and was lower for image-based items [10]. In an exploratory test of ten equine vignettes, GPT-4o's most probable diagnosis was usually accurate, but urgency was sometimes rated higher than by a specialist [11]. Hallucination (confident but implausible responses), undisclosed training data, privacy and absent regulatory oversight are repeatedly flagged [9].

Clinical documentation. Documentation is the leading reported AI use (62.8% of AI users in the 2026 survey) [49]. Efficiency figures come from vendors: a Scribenote case study (67 doctors, four weeks) reports 70 minutes saved per doctor per day and more than 647,000 SOAP notes [57]; an Australian report describes up to 9 hours per week saved with the Heidi Health scribe [58]; IDEXX reports 45% fewer clicks in its Web PACS and 10-minute results from inVue Dx [60]. All are company-reported [V]; we found no independent peer-reviewed time-motion study.

3.6 Oncology tools and livestock analytics

A heat-diffusion scanner (HT VET) evaluated on 1,010 masses in 669 dogs ruled out malignancy with 92% sensitivity and 98% negative predictive value, and the company reports use in more than 600 clinics [35]. A deep-learning system reading polarisation-sensitive optical coherence tomography of excised tumour margins reached AUC 0.989 and 91% accuracy within minutes of surgery [36]. In livestock, neural networks predicted subclinical mastitis with 84% accuracy from 364,249 milking records, a GRU autoencoder flagged respiratory disease in pigs from environmental sensors (precision and recall 0.909), and a knowledge-graph model classified dairy-cow disease with F1 above 0.85 from 21,649 records [1]. The review judged disease diagnosis from sequential sensor data immature compared with behaviour tracking [1], and only 4 of 39 studies covered livestock or horses despite their economic and public-health weight. Commercial activity is growing, for example a cattle feed-recommendation system and the acquisition of a dairy-monitoring company in 2024 [56].

Table 2. Reported performance in pathology, health-record, oncology, livestock and language-model studies (as reported; not pooled).

Task	Data	Reported result	Type	Ref.
Feline reticulocyte classification	1,046 smears	Accuracy 98.7%	P	[1]
Equine EIPH cytology	52 slides	AI 92.3% vs human experts 76%	P	[1]
Canine skin tumours (7 classes), earlier study	350 slides	About 95% (below human experts)	P	[1]
Canine skin tumours (7 classes), 2026 proof-of-concept	Whole-slide images	Accuracy 75.6-80.9%	P	[26]
Canine lymphoma (3 classes)	1,530 WSI	Accuracy 99% (preliminary)	P	[1]
Canine lymphoma vs reactive hyperplasia	Cytology images	Acc 95.4%; AUC 0.988	P	[33]
Urine sediment analyser (SediVue Dx)	530 samples	Sens 85-90%; spec 87-90%	P	[32]
Feline CKD early-risk prediction	106,251 records	Sens 0.907; spec 0.989	P	[1]
GPT-4o sign extraction, feline EHR	250 records	Sens 96.9%; spec 97.6%; PPV 80.7%	P	[8]
Necropsy report classification (LSTM)	1,000 reports	F1 0.932 / 0.947 / 0.752 (GI / resp / urinary)	P	[1]
Canine mass screening (heat diffusion)	1,010 masses, 669 dogs	Sens 92%; NPV 98%	P (company-linked)	[35]
Tumour-margin assessment (PS-OCT)	Excised margins	AUC 0.989; acc 91%	P	[36]
Subclinical mastitis, dairy cows	364,249 records	Accuracy 84%	P	[1]
Respiratory disease, pigs (sensors)	Farm sensor data	Precision 0.909; recall 0.909	P	[1]
Veterinary exam questions, GPT-3.5 / GPT-4	495 questions	55% / 77% vs students 86%	P	[63,9]

4. From Laboratory to Clinic: The Real-World Evidence Gap

The laboratory figures above describe what algorithms can do on curated data. This section assembles the independent evaluations that describe what commercial tools do in practice, and places both in one frame (Table 3).

4.1 Abdominal and thoracic radiography

In an independent pilot study (JAVMA 2026), six commercial veterinary radiology AI platforms were tested on canine abdominal radiographs from general practices with confirmed diagnoses. Performance was described as variable and mostly low to moderate: mean accuracy 70-90%, balanced accuracy 60-65%, sensitivity 28-78%, F1 28-51% and PPV 25-54%. Sensitivity for small-intestinal obstruction, a surgical emergency, was only 23-69% [23]. The authors noted that vendor claims rest largely on thoracic cases, internal validation or controlled conditions, and that for some platforms only marketing claims are publicly available [23]. In a separate independent evaluation against CT-confirmed findings in 56 dogs, a commercial tool detected pulmonary nodules and masses with 69.3% accuracy, 55.4% sensitivity and 93.7% specificity [27], so roughly 45% of confirmed lesions were missed.

4.2 Devices and triage

In 105 companion animals (54 dogs, 51 cats), an AI-enabled digital stethoscope effectively failed to detect murmurs in cats, labelled all canine rhythms as abnormal (0% specificity for absence of arrhythmia), and its atrial-fibrillation calls had a PPV of 21.4% (22 false positives versus 6 true cases). Fourth-year students matched its murmur detection, and experienced clinicians outperformed it [24]. In emergency triage, AI models correctly identified 80-90% of critical cases but flagged about 60% of non-urgent cases as urgent [25].

4.3 Human-AI interaction runs in both directions

The source review interpreted the finding that veterinarians shown AI output did not improve as likely scepticism among experts [1,4]. The stethoscope study documents the opposite risk: veterinarians began second-guessing their own judgements on the strength of the device output [24], and human-medicine evidence indicates that faulty AI can reduce clinician accuracy [39]. Both directions are plausible and neither has been measured systematically in veterinary practice; the relevant question is whether the clinician-plus-AI system outperforms the clinician alone, which most studies do not test [4,39].

Table 3. Laboratory versus independent real-world evidence (as reported; tasks differ and are not pooled).

Domain	Laboratory / research result [P]	Independent / real-world result [I]	Ref.
Radiography	LAE detection acc 82.71%, sens 68.42% (792 images); tracheal collapse acc 98.9%	Six commercial platforms, canine abdomen in general practice: sens 28-78%, PPV 25-54%; obstruction sens 23-69%	[7,1,23]
Thoracic nodules	Cardiomegaly AUROC above 0.9; lesion classification error 10.7%	Commercial tool vs CT: acc 69.3%, sens 55.4%, spec 93.7% (56 dogs)	[6,4,27]
Commercial vs specialist	Matched best of 11 radiologists in accuracy; less sensitive (50 cases)	Design criticised: small, single-site, no gold standard	[2,3]
Cardiac auscultation	Not applicable	AI stethoscope: murmurs missed in cats; AF PPV 21.4%	[24]

Domain	Laboratory / research result [P]	Independent / real-world result [I]	Ref.
Triage	Not applicable	80-90% critical cases found; about 60% non-urgent over-triaged	[25]
Skin-tumour cytology	About 95% (earlier study)	75.6-80.9% on whole-slide images; authors warn of overestimation	[1,26]
EHR extraction	Sens 96.9%, spec 97.6%	PPV 80.7% at low prevalence (single hospital)	[8]
Documentation tools	Not applicable	Vendor-reported time savings only [V]; no independent time-motion study	[57,58,60]

The pattern is consistent: high sensitivity and accuracy are reported where labels are reliable and data resemble the training set, and performance degrades on heterogeneous general-practice data, on rare classes and at low prevalence. Importantly, the independent evaluations concern different tasks from the laboratory studies, so Table 3 shows the direction and size of the problem, not a paired decline for any single model.

5. Appraisal of Benefits and Harms

5.1 Benefits supported by the evidence

(a) **Consistency and speed in narrow tasks.** Where the target is well defined and labels reliable, models match or exceed human readers: vertebral heart scale agreement (ICC 0.998) [5], equine cytology (92.3% vs 76%) [1], thoracic lesion classification (10.7% error) [4], and lymphoma cytology (AUC 0.988) [33]. (b) **Access to scarce expertise.** Board-certified radiologists are scarce; AI that matches the best radiologist on specificity, even with lower sensitivity, may be useful as a triage or second-reader layer [2]. AI radiology reports are delivered within minutes compared with same-day to 48 hours for teleradiology [59] [V]. (c) **Standardisation.** Algorithmic mitotic counting avoids variability from differing slide-region choice [1]. (d) **Rapid intraoperative information.** Margin assessment within minutes of surgery rather than days awaiting histopathology [36]. (e) **Scalable surveillance.** Necropsy-report classification and EHR extraction enable syndromic surveillance and retrospective research beyond manual capacity [1,8]. (f) **Herd-level early warning.** Sensor and milking models allow continuous monitoring [1]. (g) **Reduced documentation burden** is plausible and widely reported but rests on vendor data only [57,58] [V].

5.2 Harms and risks

Table 4. Principal harms, supporting evidence and mitigation.

Risk	Evidence from the literature	Mitigation
Data scarcity and imbalance	52.6% of studies under 1,000 cases (this analysis); rare classes failed (neoplasm sens 0-37.5%; hip dysplasia sens 0.53) [1]	Multi-centre datasets; report class prevalence and per-class sensitivity
Species and breed fragmentation	82.1% dogs/cats; 4 livestock or equine studies [1]; a model trained on small breeds may underperform where large breeds predominate [34]	Species- and breed-specific validation; international consortium
Weak or absent independent validation	Internal validation does not capture clinical variation [1]; vendor claims rest on internal or thoracic data [23]; nearly two-thirds of 71 vendors disclosed no validation metric [21]	External and prospective validation; transparent reporting
Real-world underperformance	Sens 28-78% (abdomen) [23]; sens 55.4% (nodules) [27]; AF PPV 21.4% [24]	Independent evaluation before purchase; failure reporting
Opacity (black box)	None of the 39 studies addressed interpretability technically [1]; reasons of deep networks may be unknowable [37]	Saliency (Grad-CAM) and LIME-type explanations; disclosure of training data
Automation bias and skill erosion	Veterinarians second-guessed own judgement [24]; faulty AI lowers accuracy [39]; 68% of BVA-survey vets worried about undermining of skill [50]; no measured incidence of skill erosion found	Human-in-the-loop design; audit of override rates
Hallucination and over-triage	LLMs below students (77% vs 86%) and weaker on images [63,10]; urgency over-estimated [11]; 60% non-urgent over-triaged [25]	Grounding in formularies and guidelines; clinician verification
Low-prevalence false positives	GPT-4o PPV 80.7% despite specificity 97.6% [8]	Report PPV and NPV with prevalence; confirm positives
Privacy and data ownership	Ownership, secondary use, training on client data, retention and cross-border storage unresolved [41]; de-identified data can be re-identified [37]	Data-governance policy; informed client consent
Dataset bias	Models can learn device fingerprints, stains, hospital income or age rather than disease [38]	Multi-site, balanced training data; bias audits
Cost and digital divide	Unlimited AI screening about USD 200/month; radiologist report from USD 86 per region; AI plus human oversight about USD 75 or less per study [61]; no cost-effectiveness study found	Independent cost-effectiveness studies; shared regional services
Regulatory gap	No FDA pre-market requirement for veterinary AI; limited post-market authority [9,43]; specialty colleges state no commercial imaging product meets standards [42]	Voluntary pre-market evaluation; professional standards

6. A Proposed Veterinary Translational Readiness Index (VTRI)

Performance numbers alone are not comparable across tasks, and a high figure may come from a small, single-site test. We therefore score each application class on five domains drawn from the limitations the literature itself

identifies (Table 5). The total (0-15) summarises how ready a class is for routine, unsupervised clinical deployment; it does not measure how accurate any model is. The RWP flag (Section 2.5) records whether independent real-world data agree with laboratory results.

Table 5. VTRI scoring rubric (0-3 per domain).

Domain	0	1	2	3
D1 Data adequacy	Under 500 cases or single class	500-2,000, limited diversity	Over 2,000, multiple sources	Large, multi-centre, curated, class-balanced
D2 Validation	None or internal only (unreported)	Internal / single-site test set	External validation (any outcome)	Prospective multi-site evaluation
D3 Interpretability	Not addressed	Post-hoc maps on a subset	Systematic explainability reporting	Explanations validated against clinician reasoning
D4 Species coverage	Single species / breed set	Dogs and cats only	Two or more species groups	Companion, livestock and equine validated
D5 Integration	Research only	Pilot or limited commercial	Commercial use or lab deployment	Regulated, monitored post-market

Table 6. Preliminary VTRI scores and RWP flags (author's judgement from the cited evidence; to be validated).

Application class	D1	D2	D3	D4	D5	VTRI /15	RWP	Basis
Thoracic radiography	2	2	0	1	2	7	Amber	Largest dataset (22,120) [4]; small head-to-head [2,3]; independent CT-referenced evaluation exists but shows missed lesions [27]
Commercial abdominal radiography	1	2	0	1	2	6	Red	Training data undisclosed; independent evaluation found sens 28-78% [23]
Cytology / histopathology	1	1	0	1	2	5	Amber	Median n = 352 [1]; large-laboratory deployment reported [1]; later study warns of overestimation [26]
EHR text / LLM tools	2	1	0	1	1	5	Amber	Large EHR cohorts [1]; single-hospital GPT-4o test [8]; exam and triage gaps [9,10,25]
Clinical documentation scribes	1	0	0	1	2	4	Amber	Widely used [49]; efficiency data vendor-reported only [57,58]
Livestock sensor analytics	2	0	0	2	1	5	Amber	Very large records [1]; disease diagnosis judged immature [1]
MRI / ultrasound / CT	1	0	0	1	0	2	Amber	Rare-class failures; poor ground truth in CKD staging [1]

Three patterns emerge. First, D3 (interpretability) scores 0 for every class, consistent with the finding that no included study addressed it [1]. Second, scores cluster between 2 and 7 of 15: even the most mature classes are less than half-way to routine unsupervised use. Third, readiness and real-world performance can diverge: commercial abdominal radiography scores 6 because it is on the market and has been externally tested, yet carries a Red flag because that external test exposed deficits. This is why the index pairs the score with the RWP flag, and why D2 rewards the existence of external validation, not its result. The index is deliberately simple; its value lies in making disagreements explicit, since any reader who rescored a cell would have to name the evidence that justifies the change.

7. Adoption, Market and Governance

7.1 Adoption and market

Adoption figures come from surveys with different populations and definitions and cannot be pooled. A survey of 3,968 veterinary professionals (December 2023 to January 2024) found 39.2% using AI, with reliability/accuracy (70.3%) and data security (53.9%) as the top concerns [48]. A 2026 survey of 1,730 professionals in the United States and Canada reported 83.7% use, with clinical documentation the leading application (62.8%) and fear of job loss down from 37.2% to 20% [49]. A British Veterinary Association survey reported that 21% of vets used AI, with concerns about interpretation without context (83%), use without manual checks (82%), undermining of human skill (68%) and lack of data protection (25%) [50]. The difference between 83.7% and 21% reflects different countries, sampling and definitions of use, not necessarily a real contrast in uptake.

Market estimates for AI in animal health differ because firms define the market differently (Table 7). All are projections or commercial estimates [F] and should be treated as indications of commercial momentum, not evidence of clinical value. North America held 38.4% of the market in 2025 (54.9% for the diagnostics segment), and diagnostics accounted for 54.5% of AI-in-animal-health revenue [51,52].

Table 7. Market estimates for AI in animal health (commercial estimates and projections; definitions differ).

Source	Estimate	Type	Ref.
Grand View Research (2026)	USD 1.8 billion in 2025; USD 7.0 billion by 2033; CAGR 18.9%	F	[51]
Grand View Research (2025), diagnostics	USD 652.2 million in 2024	F	[52]
SNS Insider (2026)	USD 1.88 billion in 2025; USD 12.8 billion by 2035; CAGR 20.94%	F	[53]

Source	Estimate	Type	Ref.
Research and Markets (2026)	USD 1.2 billion (2024); USD 1.44 billion (2025); USD 2.88 billion by 2029	F	[54]
The Business Research Company (2025), diagnostics	More than USD 4.05 billion by 2029; CAGR 20.2%	F	[55]
Roots Analysis	USD 2.00 billion in 2026; USD 8.87 billion by 2035; CAGR 18.0%	F	[56]

7.2 Regulation and professional position

The American College of Veterinary Radiology and the European College of Veterinary Diagnostic Imaging (2025) stated that no commercial AI product for veterinary diagnostic imaging currently meets the required standards of transparency, validation or safety, and called for veterinarian-in-the-loop use, post-deployment monitoring and disclosure of AI use to owners [42]. The American Association of Veterinary State Boards (March 2025) obliged licensees to understand AI risks and limitations, and noted that the FDA lacks legal authority to require pre-market notification for animal devices, its oversight being post-market [43]. The British Veterinary Association (December 2025) issued eight principles calling for active regulation and transparent validation data [44]. The Federation of Veterinarians of Europe has had an AI working group since 2019 and includes technology in its 2026-2030 strategy, but no binding position was found [45]. Canadian guidance is also available [46], and practice-level governance policies address over-reliance and skill atrophy [47]. These statements reflect 2024-2026 sources and the regulatory landscape is changing quickly.

8. Discussion

8.1 When AI succeeds and when it fails

The combined evidence shows a clear pattern. AI succeeds in well-defined tasks with reliable labels and a clear gold standard: vertebral heart scale, equine and lymphoma cytology, thoracic lesion classification, margin assessment [4,5,33,36]. It falters on heterogeneous, open-ended general-practice tasks, rare classes and low-prevalence findings: abdominal radiographs, pulmonary nodules, murmurs in cats, atrial fibrillation, rare MRI classes [1,23,24,27]. Deep networks are good at pattern matching within the training distribution but degrade with different equipment, non-standard conditions and breed diversity, and most declared validation is internal [1,23,38].

8.2 The adoption paradox

Reported use has risen sharply while the specialty colleges state that no commercial imaging product meets validation standards [42,48,49]. The leading drivers of adoption (documentation, speed) are the least clinically risky, while the highest-risk applications (independent diagnosis) are the least validated [49,23]. The implication is that much current use occurs in a validation and regulatory gap, even if the highest-risk uses are, for now, not the most common ones.

8.3 Overtrust, undertrust and the unit of evaluation

The central question is not whether the algorithm is accurate but whether the clinician-plus-AI system is more accurate than the clinician alone. Veterinary data point both ways: AI output did not improve veterinarians in thoracic radiology [4], and clinicians second-guessed correct judgements because of a faulty stethoscope [24]. A veterinary evidence base for human-AI performance, with measured override rates, is a priority.

8.4 The global gap: a perspective from Algeria and other low-resource settings

Evidence, products and investment are concentrated in North America [51,52], and we found no quantitative data on the effect of veterinary AI in developing countries. This poses a dual challenge. Low-cost tools could help address shortages of radiology and pathology specialists in countries such as Algeria; yet models trained on northern breeds, equipment and case mix may transfer poorly [34], and subscription prices of the order of USD 200 per month [61] may deepen the divide. Both statements are reasoned inferences from the sources, not findings. Imported tools should therefore be validated locally, on local breeds, equipment and disease prevalence, before routine use, and local data contribution to a multi-species consortium is a practical route to better evidence [1].

8.5 Recommendations

Table 8. Priorities by audience and horizon.

Audience	Priority	Rationale
Practising veterinarians	Use AI as decision support only; keep the veterinarian in the loop; ask vendors for independent external validation before purchase; disclose AI use to owners; document and report tool failures	Real-world sensitivity may be 28-78% [23]; colleges require oversight and disclosure [42]
Professional and regulatory bodies (including Arab veterinary bodies)	National or regional standards for validating tools before marketing; position statements on liability and record requirements; local validation programmes for imported tools	Regulation lags adoption [42-44]; transfer to new populations is untested [34]

Audience	Priority	Rationale
Researchers (near term, 2026-2028)	External and prospective validation on general-practice data; per-class and prevalence-aware reporting; explainability reporting; independent time-motion and cost-effectiveness studies	Internal-only validation and opacity are the commonest gaps [1]; efficiency claims are vendor-only [57,58]
Researchers (mid term, 2028-2031)	Multi-species data consortium and shared benchmarks; transfer learning, vision transformers and synthetic data for scarce labels; veterinary-specific, grounded LLMs	Dataset size has not grown [1]; general LLMs trail students [9,10]
Researchers (longer term, 2031+)	Multimodal models combining images, records and sensors; prospective randomised evaluation of clinical outcomes	Multimodal models exist in human medicine [1]; no outcome trials were identified in the reviewed sources
Developers and companies	Publish validation data including failure cases and generalisation limits; design systems with uncertainty warnings that support, not replace, judgement	Two-thirds of vendors disclosed no validation metric [21]

The near- and longer-term priorities above are the author's synthesis of directions proposed in the sources [1,9,23,42] and are forecasts, not findings.

9. Limitations

This is not a new systematic review. Its backbone is a single published PRISMA review whose search used two keywords in one database, so non-deep-learning machine learning, non-PubMed literature and 2025-2026 work may be missed. Secondary extraction was done by one analyst from a published table, and sample-size units are heterogeneous. Statistical tests are exploratory. Headline metrics were not pooled and cannot be compared across tasks, and the laboratory and real-world studies in Table 3 concern different tasks. Several LLM results come from small or exploratory studies [10,11]. Some figures derive from company, press or secondary reports rather than the primary paper [20,21,35,36,50,57-61], and the VATI audit [21] was known to us only through a secondary report. Regulatory statements and market projections reflect the situation at the time of writing. The novelty search was targeted, not systematic. VTRI scores and RWP flags are subjective, single-rater and unvalidated. Future work should (i) run a formal multi-database systematic search, (ii) have two independent reviewers extract data, and (iii) validate VTRI through a modified Delphi study with veterinary radiologists, pathologists, clinicians and data scientists, reporting inter-rater agreement.

10. Conclusions

AI in veterinary medicine delivers real benefit in narrow, well-labelled tasks, from heart-size measurement to cytology and intraoperative margin assessment, and offers scalable surveillance from text and sensor data. Yet more than half of the 39 deep-learning studies rest on fewer than 1,000 cases, over four-fifths concern dogs and cats, none addressed interpretability, and the few independent evaluations of commercial tools show sensitivity far below laboratory figures. Readiness to deploy is therefore a different question from accuracy, and the VTRI, paired with the Real-World Performance flag, offers a transparent way to ask it. Pending prospective, multi-site, multi-species evidence, veterinary AI should be deployed as clinician-supervised decision support, with independent external and local validation, prevalence-aware reporting and explicit accountability.

Declarations

Use of AI tools. The author used a large language model (Claude, Anthropic) to assist with literature organisation, calculations and drafting of text. The tool is not an author. The author designed the study, verified the sources, figures and calculations, and takes full responsibility for the content.

Ethics. No animals or human participants were used.

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Competing interests. The author is the founder of an AI-based veterinary service (vetai.vcom.workers.dev). This service is not evaluated or promoted in this manuscript. The author has no financial relationship with any company whose products are discussed in this manuscript and declares no other competing interests.

Data availability. Appendix A lists the extracted variables for all 39 studies; the original data are in Table 1 of Xiao et al. [1].

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Appendix A. Extracted variables for the 39 studies (from Table 1 of Xiao et al. [1])

#	Year	Modality group	Species	Reported sample size
1	2013	Radiograph	Canine	256
2	2018	MRI	Canine	80
3	2018	Ultrasound	Canine	52
4	2018	Cytology/Histology	Feline	1,046
5	2018	Radiograph	Canine	7,138
6	2018	Sensor/EHR/Text	Swine	not specified
7	2019	Sensor/EHR/Text	Feline	106,251
8	2019	Photo/Video	Canine	281
9	2019	Cytology/Histology	Canine	1,500
10	2019	Radiograph	Canine/Feline	2,862
11	2019	Sensor/EHR/Text	Bovine	364,249
12	2020	Radiograph	Canine	792
13	2020	Cytology/Histology	Canine	32
14	2020	Sensor/EHR/Text	Unspecified	1,000
15	2020	Radiograph	Canine	1,465
16	2020	Cytology/Histology	Canine	352
17	2020	Radiograph	Canine/Feline	22,120
18	2021	MRI	Canine	2,693
19	2021	Cytology/Histology	Canine	416
20	2021	Cytology/Histology	Rodent	300
21	2021	Cytology/Histology	Canine/Feline	3,845
22	2021	Radiograph	Canine	2,274
23	2021	Radiograph	Canine/Feline	60
24	2021	Sensor/EHR/Text	Bovine	21,649
25	2022	CT	Canine	386
26	2022	Photo/Video	Canine	95
27	2022	Radiograph	Feline	500
28	2022	Cytology/Histology	Equine	52
29	2022	Radiograph	Canine	1,000
30	2023	Radiograph	Feline	273
31	2023	Radiograph	Canine	2,382
32	2023	Cytology/Histology	Canine	1,056
33	2023	Cytology/Histology	Canine	350
34	2023	Cytology/Histology	Rodent	33
35	2023	Cytology/Histology	Canine/Feline	154
36	2023	MRI	Canine	5,991
37	2024	Cytology/Histology	Canine	1,530
38	2024	Radiograph	Canine	600
39	2024	Ultrasound	Canine	883

Modality grouping: radiograph; cytology/histology (microscope slides, cytology and histopathology images); MRI; ultrasound; CT; photo/video; sensor/EHR/text (health records, necropsy reports, sensor data). Mixed-species studies are listed as Canine/Feline. Sample units vary (images, animals, slides, records).